SE LAB 5 -> 39-vehicle service

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Sec: H

Initial setup:

```
PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m venv venv
PS C:\Users\Shre^Cakshmi Bhat\Desktop\Projects\L5_39_vehicle_service>
PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pip install --upgrade pip
```

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pip install --upgrade pip
>>
Requirement already satisfied: pip in c:\users\shreelakshmi bhat\desktop\projects\l5_39_vehicle_service\venv\lib\site-p
ackages (24.0)
Collecting pip
  Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
Successfully installed pip-26.2.1
```

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pip install pylint
Collecting pylint
  Using cached pylint-4.1.3-py3-none-any.whl.metadata (12 kB)
Collecting astroid<=4.4,>=4.3.4 (from pylint)
  Using cached astroid-4.3.4-py3-none-any.whl.metadata (4.4 kB)
Collecting colorama>=0.4.5 (from pylint)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting dill>=0.3.6 (from pylint)
  Using cached dill-0.4.1-py3-none-any.whl.metadata (10 kB)
Collecting isort!=5.13,<10,>=5 (from pylint)
  Using cached isort-9.0.2-cp311-cp311-win_amd64.whl.metadata (11 kB)
Collecting mccabe<0.8,>=0.6 (from pylint)
  Using cached mccabe-0.7.0-py2.py3-none-any.whl.metadata (5.0 kB)
Collecting platformdirs>=2.2 (from pylint)
```

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pip install flake8
Collecting flake8
  Using cached flake8-7.4.1-py2.py3-none-any.whl.metadata (3.8 kB)
Requirement already satisfied: mccabe<0.8.0,>=0.7.0 in .\venv\Lib\site-packages (from flake8) (0.7.0)
Collecting pycodestyle<2.16.0,>=2.15.0 (from flake8)
  Using cached pycodestyle-2.15.0-py2.py3-none-any.whl.metadata (4.5 kB)
Collecting pyflakes<4.1.0,>=4.0.0 (from flake8)
  Using cached pyflakes-4.0.3-py2.py3-none-any.whl.metadata (3.7 kB)
Using cached flake8-7.4.1-py2.py3-none-any.whl (57 kB)
Using cached pycodestyle-2.15.0-py2.py3-none-any.whl (31 kB)
Using cached pyflakes-4.0.3-py2.py3-none-any.whl (66 kB)
Installing collected packages: pyflakes, pycodestyle, flake8
```

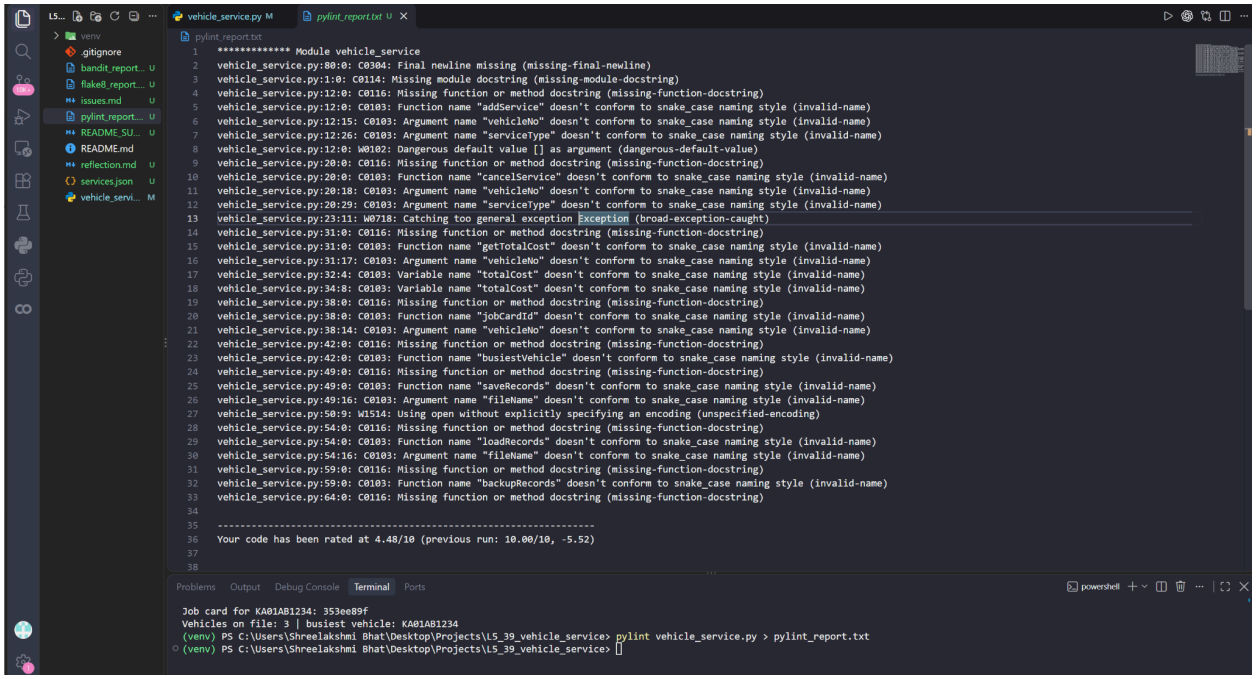
```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pip install bandit
Collecting bandit
  Using cached bandit-1.9.4-py3-none-any.whl.metadata (7.1 kB)
Collecting PyYAML>=5.3.1 (from bandit)
  Using cached pyyaml-6.0.3-cp311-cp311-win_amd64.whl.metadata (2.4 kB)
Collecting stevedore>=1.20.0 (from bandit)
  Using cached stevedore-5.9.1-py3-none-any.whl.metadata (2.3 kB)
Collecting rich (from bandit)
  Using cached rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Requirement already satisfied: colorama>=0.3.9 in .\venv\Lib\site-packages (from bandit) (0.4.6)
Collecting markdown-it-py>=2.2.0 (from rich->bandit)
```

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m pylint --version
pylint 4.1.3
astroid 4.3.4
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)]
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m bandit --version
__main__.py 1.9.4
  python version = 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 10:12:12) [MSC v.1938 64 bit (AMD64)]
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m flake8 --version
7.4.1 (mccabe: 0.7.0, pycodestyle: 2.15.0, pyflakes: 4.0.3) CPython 3.11.9 on Windows
```

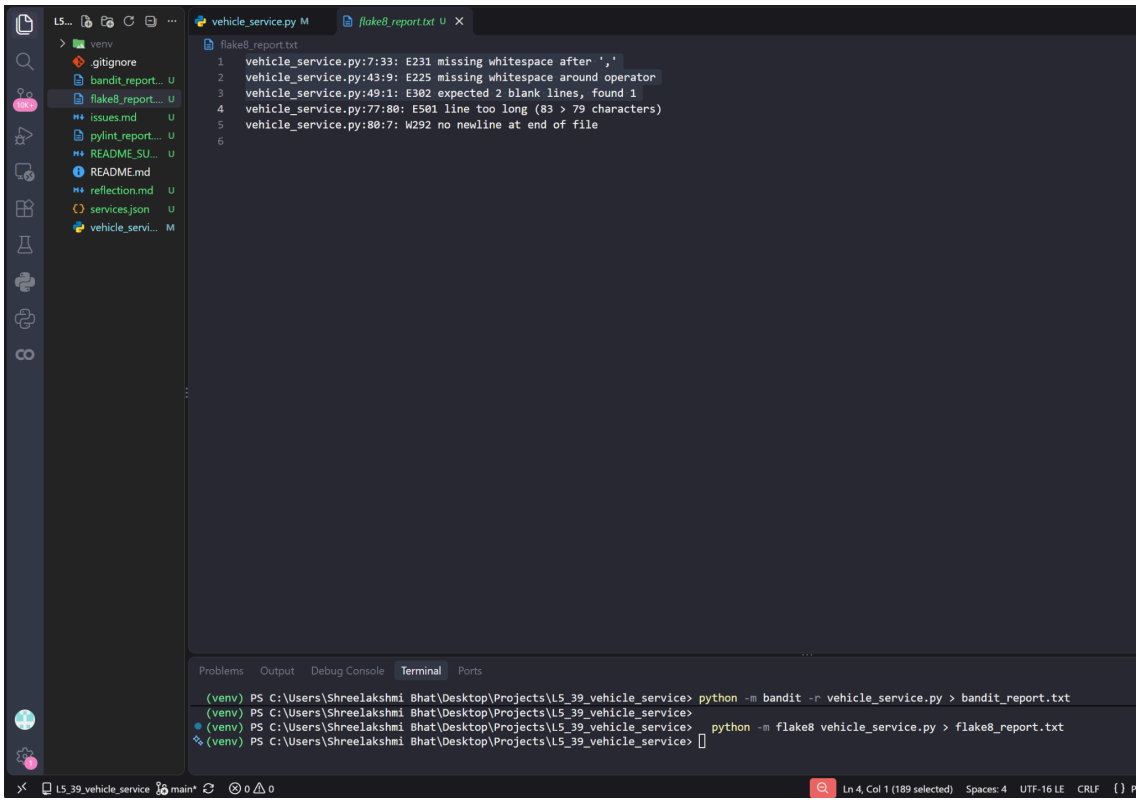
Before correcting the errors in vehicle_service.py:

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python vehicle_service.py
No records for KA09ZZ0001
Total for KA01AB1234: 2300
Total for KA05XY9876: 2500
Job card for KA01AB1234: 353ee89f
Vehicles on file: 3 | busiest vehicle: KA01AB1234
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service>
```

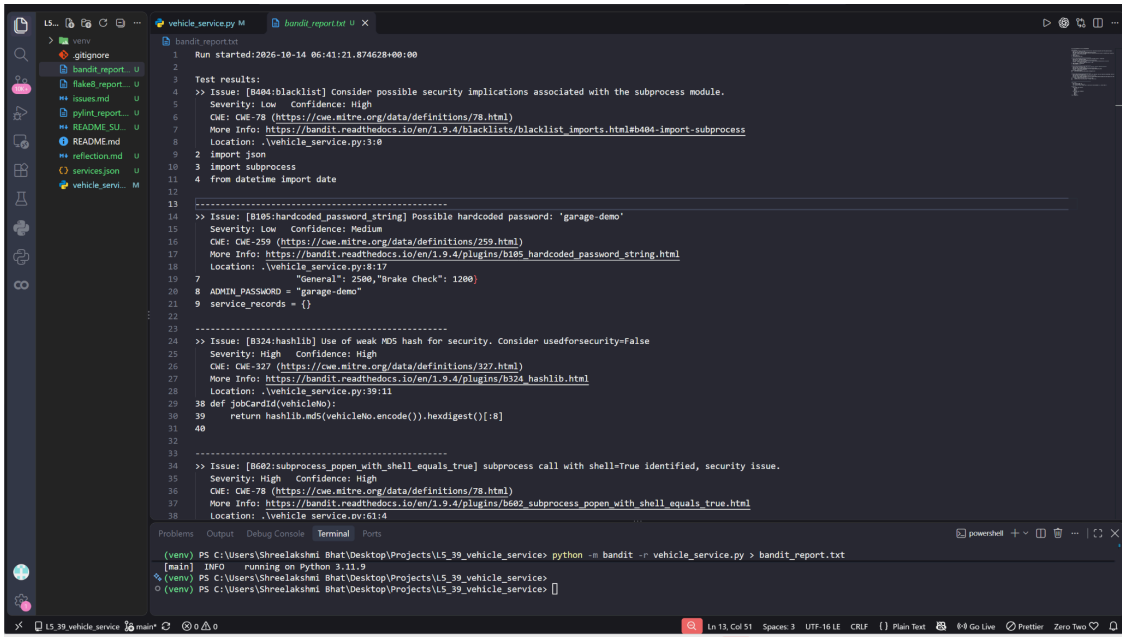
before-> pylint



before->flake8



before->bandit



```
Run started:2026-10-14 06:41:21.874628+00:00

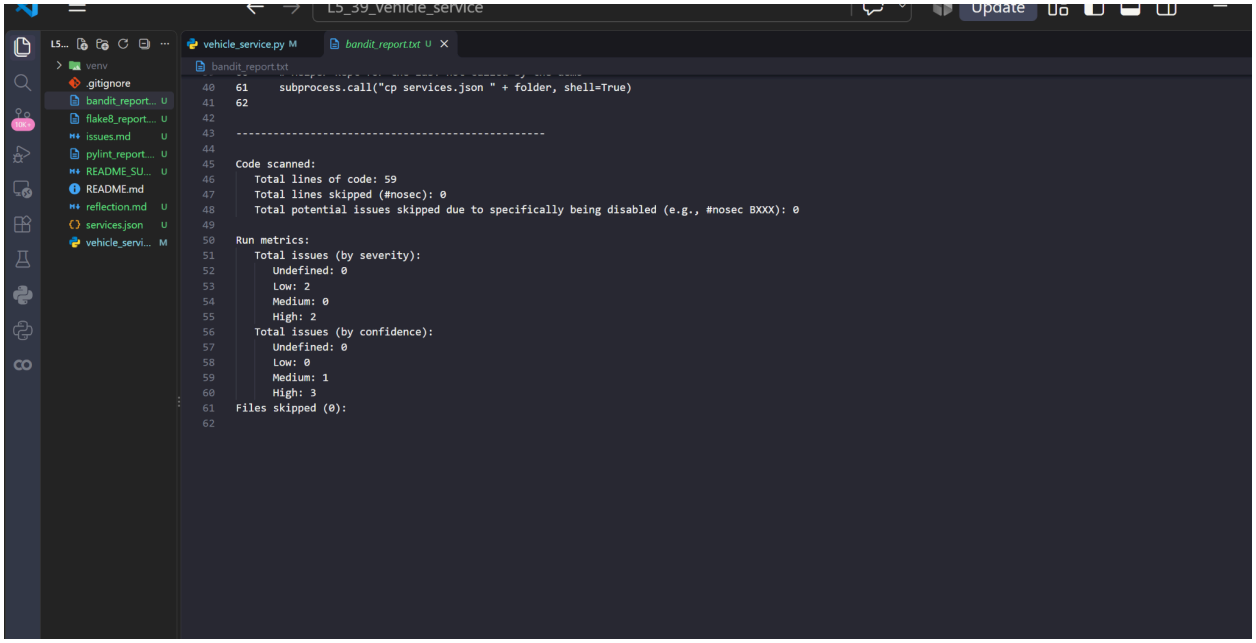
Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: .\vehicle_service.py:3:0
2  import json
3  import subprocess
4  from datetime import date

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'garage-demo'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\vehicle_service.py:8:17
7                  "General": 2500,"Brake Check": 1200}
8  ADMIN_PASSWORD = "garage-demo"
9  service_records = {}

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\vehicle_service.py:39:11
38 def jobCardId(vehicleNo):
39     return hashlib.md5(vehicleNo.encode()).hexdigest()[:8]
40

--------------------------------------------------
>> Issue: [B602:subprocess_popen_with_shell_equals_true] subprocess call with shell=True identified, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b602_subprocess_popen_with_shell_equals_true.html
   Location: .\vehicle_service.py:61:4
```

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python -m bandit -r vehicle_service.py > bandit_report.txt
[main]  INFO    running on Python 3.11.9
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service>
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service>
```



```
    subprocess.call("cp services.json " + folder, shell=True)


--------------------------------------------------

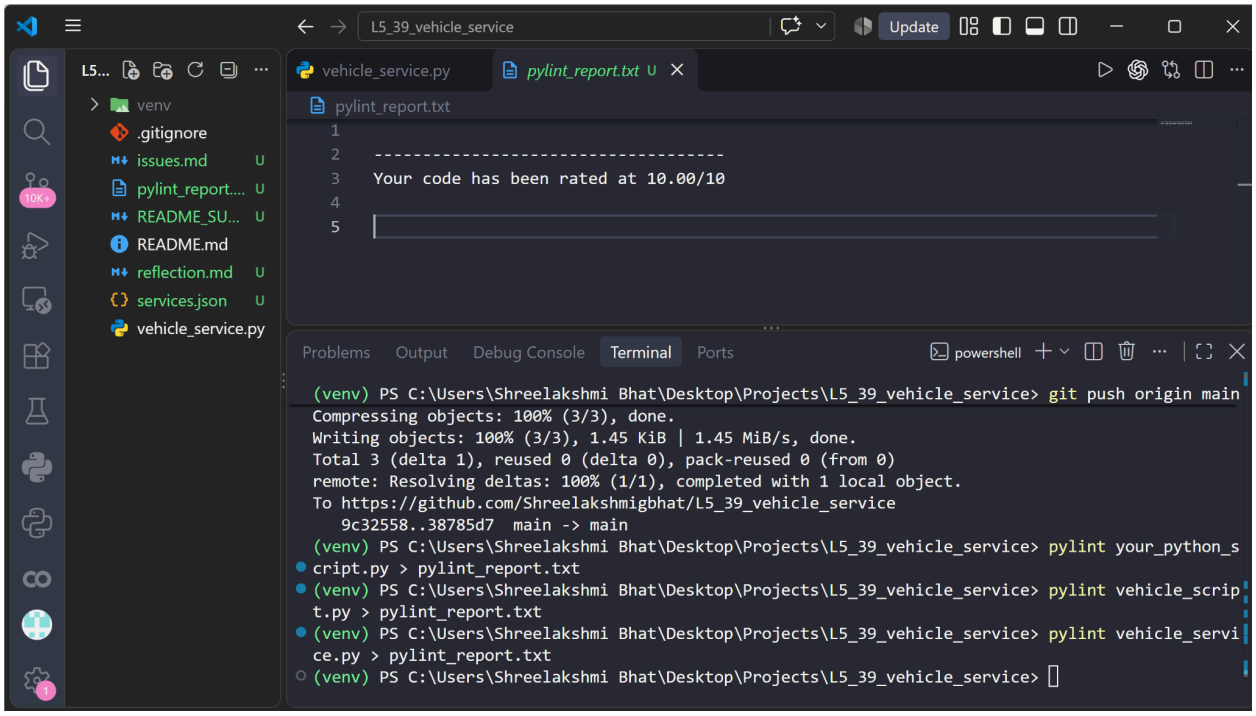
Code scanned:
    Total lines of code: 59
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 2
        Medium: 0
        High: 2
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 3
Files skipped (0):
```

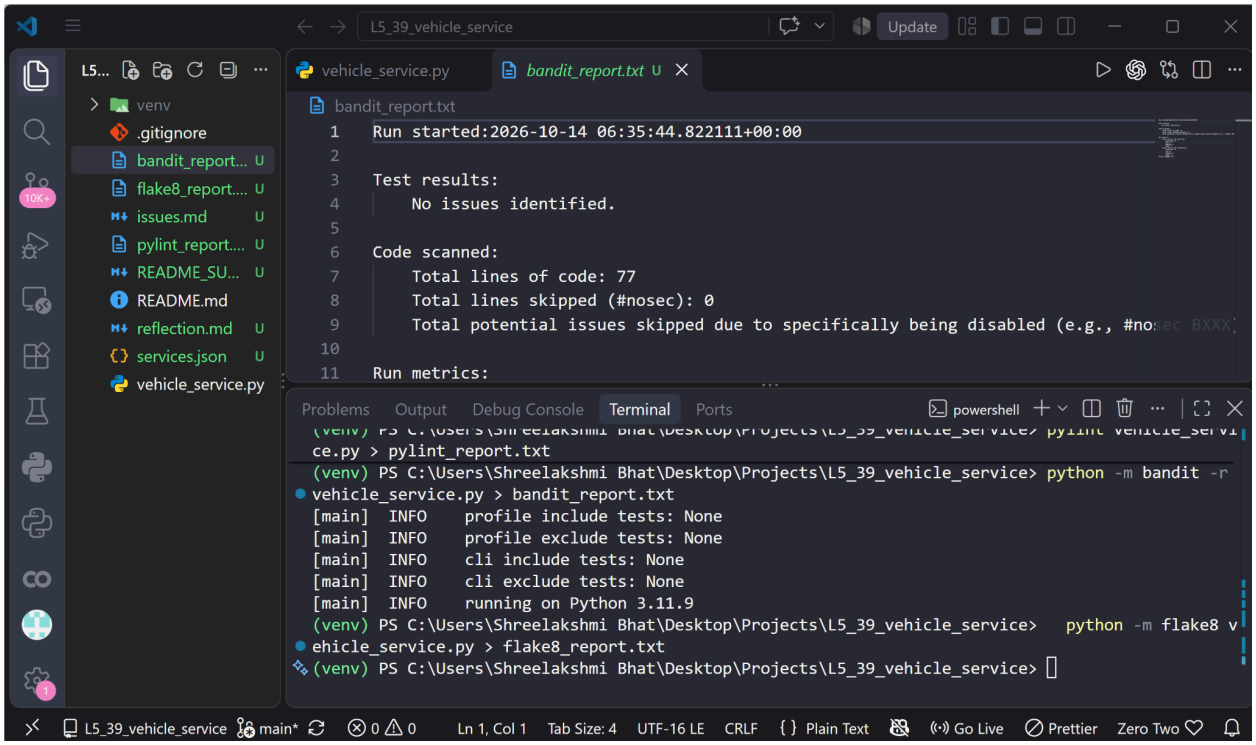
After correcting the errors in vehicle_service.py:

```
(venv) PS C:\Users\Shreelakshmi Bhat\Desktop\Projects\L5_39_vehicle_service> python vehicle_service.py
No records for KA09ZZ0001
Total for KA01AB1234: 2300
Total for KA05XY9876: 2500
Job card for KA01AB1234: bd2cd41f
Vehicles on file: 3 | busiest vehicle: KA01AB1234
```

after->pylint



after->bandit



after->flake8

